



01

Publications Brain

What to publish, what to stop, and keeping every output consistent.

Hackathon vertical 01 of 04 · build afternoon agenda, swarm, gates, question bank and impact worksheet

01 Publications Brain

02 Insights Engine

03 Congress Monitor

04 Field Intelligence Engine

AI in Action for Medical Affairs

October 13–14, 2026 · Convene Philadelphia (2nd floor), Two Commerce Square, 2001 Market St

Host and keynote: Vivek Mukhatyar · aiinaction.up.railway.app/hackathon



Scan for the hackathon pack

The vertical on one page

THE CHALLENGE

Our publication plan is sequenced by when data arrive, not by what clinicians and payers are trying to decide. And every new output (manuscript, congress abstract, poster, plain-language summary) re-states the same results by hand, so numbers and wording drift between them.

DAY 2 SLOT · WED OCT 14

9:40–10:00 AM

About 20 minutes, give or take, including a few questions. Teams present 01 → 04 from 9:40.

THE PROBLEM PROMPT

Build an AI workforce for the publications team that (1) works backwards from what each audience must decide and recommends what to publish, what to stop and what to research instead, keeping evidence gaps (nobody has answered the question: research) separate from communication gaps (it has been answered and nobody heard: publish); and (2) keeps every scientific output consistent with one source of truth, logging every discrepancy for the authors to resolve. Humans own authorship, priorities and sign-off.

Teams can keep this prompt or narrow it to one workflow they know. A good narrowed prompt names the role, the trigger, the deliverable and the decision it serves.

DONE LOOKS LIKE · DAY 1

The swarm ran end to end at least once on practice data; it stopped at Gate 1 and Gate 2 and the team made both decisions; screenshots and minute-level timings are in the capture log; the agent has drafted a slide outline.

DONE LOOKS LIKE · DAY 2

The deck is built from the final template: no [brackets] left, impact numbers are the team's own, practice data marked fictional, the LIVE DEMO slide left empty, backup screenshots in place. The team presents at 9:40 for about 20 minutes with a short live demo.

LEAD SKILL

scientific-communication-strategy

SUPPORTING SKILLS

evidence-gap-analysis

evidence-synthesis

literature-surveillance

DATA PACK

Oncology (NORVANTIB, multiple myeloma) is recommended because the Source-correction change card is written for it; any of the three works.

publication-plan.md evidence-landscape.md

field-observations.csv medical-plan.md

product-profile.md + practice CRM

Fictional practice data (Nordvant Biopharma), or bring your own non-confidential data.

WARM-UP MISSIONS

L1 medical-information

Handle the enquiry queue

L2 patient-partnership

Bring patient priorities into the plan



The build afternoon, block by block

A suggested, optional flow: **Grok Bot setup (30 min)** → **warm-up missions** → **workflow ideation** → **the hack**. Teams may move faster or slower. **5:00 PM: no formal share-out**; facilitators circulate and ask questions (page 6), then the cocktail reception.

1 · Setup 30 min	2 · Warm-ups 40 min	3 · Ideation 45 min	4 · The hack 80 min
1:45	2:15	2:55	3:40
5:00			
1 Set up Grok Bot + skills 1:45–2:15 · 30 min	2 Warm-up missions 2:15–2:55 · 40 min	3 Workflow ideation 2:55–3:40 · 45 min	4 The hack 3:40–5:00 · 80 min
1:45 Huddle 1:50 Everyone opens aiinaction.up.railway.app/grokbob 2:02 The driver pastes the setup prompt below 2:10 Smoke test passes	2:15 Split into two pairs 2:18 Both pairs run their mission in parallel 2:40 Push on it 2:50 Swap	2:55 Map today's workflow for one real publication cycle 3:10 Spot the AI ideas 3:22 Pick one problem 3:32 Sketch the swarm on paper	3:40 Write the assignment 3:50 Run 1 4:25 Optional stretch break 4:30 Run 2 4:50 Capture
DONE LOOKS LIKE Every member signed in; the driver's conversation has the skills loaded and has read one practice file; capture log started	DONE LOOKS LIKE Pair A has an enquiry triage register; Pair B has a lay summary with a list of simplifications; both seen in 2-minute swaps; 3 observations logged	DONE LOOKS LIKE The publication cycle mapped; one problem chosen (strategy or consistency); hours per cycle and outputs per year estimated; ideas marked Efficiency or Opportunity	DONE LOOKS LIKE The swarm produced a gap table and at least one draft output plus a discrepancy log, stopping at both gates; screenshots and timings captured
FACILITATORS CIRCULATE Check every laptop is signed in and the smoke test shows a file marked SYNTHETIC. GitHub blocked? Point to the starter bundle on /grokbob.	FACILITATORS CIRCULATE Let the missions finish without steering. Watch for the safety stop and the human-review moment; keep the 2-minute swap on time.	FACILITATORS CIRCULATE Ask the ideation questions (page 6). Push for one problem statement and a baseline estimate before they sketch the swarm.	FACILITATORS CIRCULATE Ask the hack, gate and data questions. From 4:45, check that screenshots and the capture log are saved.

DAY 2 · WED OCT 14 · FINAL BUILD → THIS TEAM'S SLOT

9:00–9:38 · Final build + rehearsal re-open → last fix, agent fills the template → team review → rehearse · MC opens 9:38	9:40 01 Publications Brain	10:00 02 Insights Engine	10:20 03 Congress Monitor	10:40 04 Field Intelligence Engine	11:00 WPP demo
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Publications Brain: 9:40–10:00 AM, about 20 minutes, give or take, including a short live demo and a few questions. Teams present 01 → 04 from 9:40. Keep all decks on one laptop and treat the Day 1 4:50–5:00 capture as essential. Team 01 presents first and has the least Day 2 time, so its deck draft should be done on Day 1.



Warm up, then pick your data

Warm-up missions Block 2 · 2:15–2:55

Level 1 · Starter

Pair A

“Handle the enquiry queue”

See what clinicians actually ask (the demand signal for publications) and how the agent ties every answer to the structured evidence table.

aiinaction.up.railway.app/missions/medical-information

Level 2 · Pair

Pair B

“Bring patient priorities into the plan”

Watch a lead skill hand off to plain-language-summary: the same evidence, rewritten for a lay audience without changing the numbers.

aiinaction.up.railway.app/missions/patient-partnership

STRETCH, ALTERNATIVES AND REFERENCES

L1 Starter

L2 Pair

L3 Team

L4 Swarm

L3 Alternative

Keep every scientific output consistent

</missions/publication>

L3 Optional

Choose what to fund

</missions/evidence-investment>

L4 Also useful

Build the whole medical launch plan

</missions/launch-plan-swarm>

BG Background

Team mission 4: the original team mission

[Medical-Affairs-Skills · workshop](#)

Data pack browse and copy links at aiinaction.up.railway.app/data

Fictional practice data, or bring your own non-confidential data.

Nordvant Biopharma and its products are fictional. Own data: no patient data, no confidential company data, no real HCP personal data.

PICK ONE TA

MM · NORVANTIB

AD · DERMALYX

Obesity · ADIPOSYN

Oncology (NORVANTIB, multiple myeloma) is recommended because the Source-correction change card is written for it; any of the three works.

TA PACK FILES synthetic/<TA>/

TA publication-plan.md

TA evidence-landscape.md

TA field-observations.csv

TA medical-plan.md

TA product-profile.md

TA manuscript-draft.md

TA structured-evidence-table.csv

TA patient-level-analysis.csv

TA abstract-poster-brief.md

TA plain-language-source.md

TA guideline-landscape.md

PRACTICE CRM synthetic/connected/

CRM content_assets.csv

CRM enquiries.csv

Stress test: the Source correction change card (oncology) corrects a fictional study result: ask the swarm to trace it into every affected draft.



Efficiency vs Opportunity, and the digital team

Efficiency AI today's work, faster

E1 Consistency check across outputs: every number and claim in the manuscript, abstract, poster and lay summary is traced to the structured evidence table, with a discrepancy log for authors.

E2 Encore and adaptation drafts: turn one approved source into abstracts that meet each congress's word limits and submission rules.

E3 Plain-language summary first draft from the approved manuscript, with a readability check and a list of every simplification made.

vs

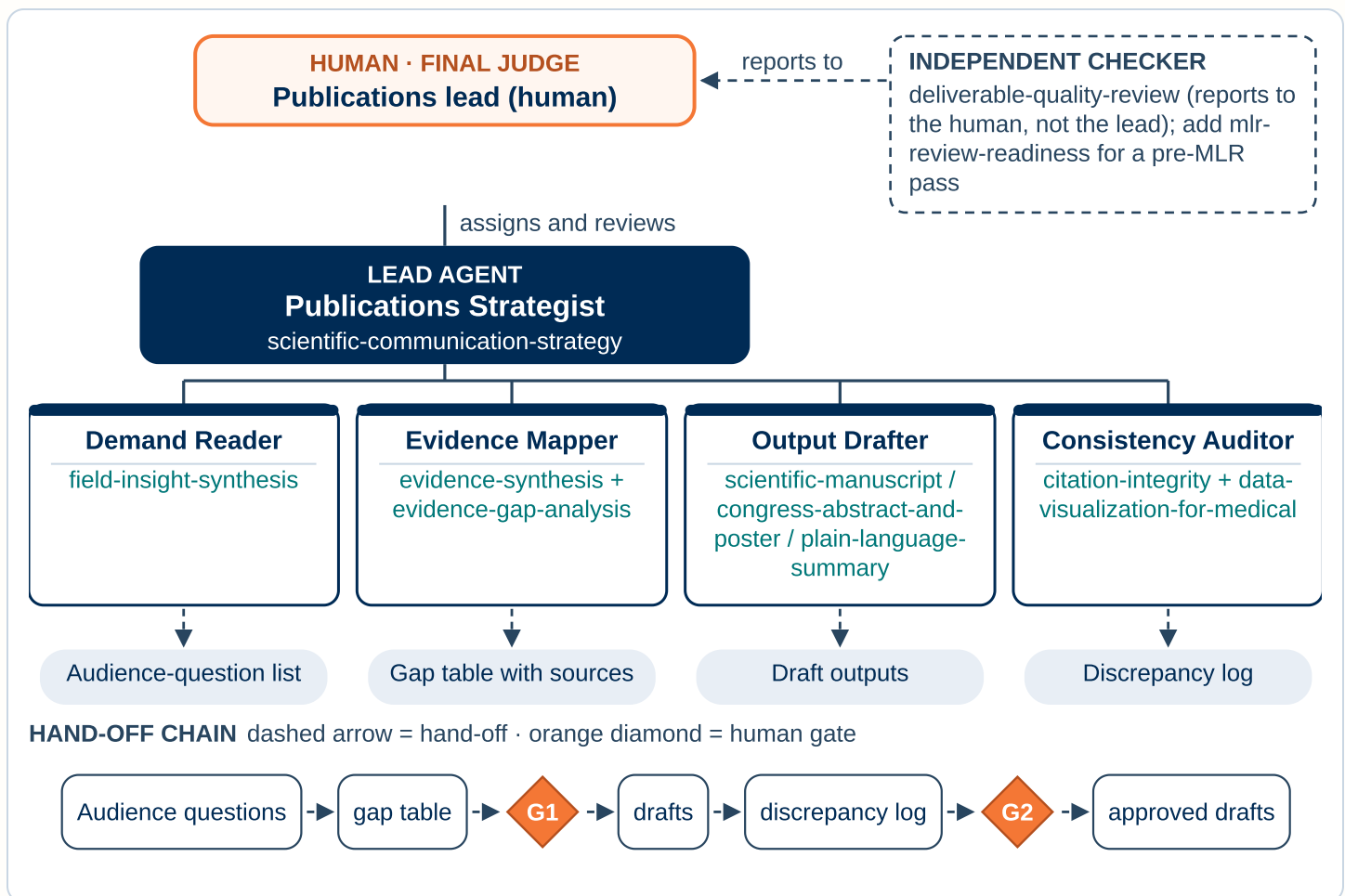
Opportunity AI something not possible today

01 A publication plan driven by what audiences must decide: map field questions and MI enquiries against the plan to show, every month, where we are silent and where we over-communicate.

02 Correction ripple: when one result changes, find and redline every affected draft, slide and published asset within minutes, instead of discovering it at MLR.

03 A living gap register that routes each gap to "research it" or "publish it", refreshed every month rather than at the annual planning cycle.

Suggested swarm one lead agent, sub-agents with their skills, two human gates



Where humans decide, and what to ask



G1

Gate 1

WHEN	After the Evidence Mapper
WHO DECIDES	The publications lead (with the steering committee)
WHAT THEY CHECK	Approves the publish / stop / research list before any drafting starts.
WHY A HUMAN	Publication priorities are a strategic and ethical decision (negative results, balance, timing).

G2

Gate 2

WHEN	After the Consistency Auditor
WHO DECIDES	Authors and the medical reviewer
WHAT THEY CHECK	Resolve the discrepancy log and approve the drafts before anything goes to authors, MLR or a journal.
WHY A HUMAN	Authorship and accountability (ICMJE), scientific judgment, no ghost-writing.

Always-on stop (a rule, not a gate): any possible adverse event, product quality complaint or off-label signal in human-sourced records is listed verbatim and routed by a human before analysis continues.

Facilitator question bank the Day 1 feedback loop: ask one or two per visit, then move on

Ideation 2:55–3:40

- 1 Who is the audience for this publication, and what decision are they trying to make with it?
- 2 Is this an evidence gap or a communication gap? How would you prove which one it is?
- 3 Which step of today's workflow takes longest, and is the delay people or approvals?
- 4 What are you publishing that nobody asked for? How would the agent know?

The hack 3:40–5:00

- 5 Which file is your single source of truth for the numbers? What happens when the agent finds a mismatch?

Gates during the hack

- 6 Where exactly does a human approve the priority list? What does the agent show them at that moment?
- 7 How does your swarm handle a negative or inconvenient result?

Data during the hack

- 8 Show me one claim in the draft and the source row it came from.
- 9 If the source-correction card landed now, how many drafts would change, and how would you know?

Impact during the hack

- 10 What would your hours-per-cycle number be for one manuscript-to-PLS cycle today? Who would sign off that estimate?



When a team gets stuck

The ten snags facilitators see most often in this vertical, each with the unblocker to offer. Nudge, then step back: the team does the building.

Pitfall	How to unblock
01 The agent lists papers in the order data arrive.	Re-ask: "Work backwards from what each audience must decide. Start with the decision, then the evidence."
02 Evidence gaps and communication gaps get mixed up.	Make the Evidence Mapper label every row "evidence gap" or "communication gap" with one line of reasoning.
03 Numbers drift in the lay summary.	Tell the Consistency Auditor to compare every number with structured-evidence-table.csv and list every difference, even rounding.
04 Trying to write a full manuscript in 80 minutes.	Scope to one output plus the discrepancy log. A short, correct draft beats a long, unchecked one.
05 Invented references.	Load citation-integrity; any claim without a source is marked "unsupported" and left for the author.
06 Authorship or ghost-writing worries in the room.	Good: make it Gate 2. The agent drafts; named authors decide, edit and own.
07 Grok Bot cannot open GitHub.	Upload the starter bundle for your TA from https://aiinaction.up.railway.app/grokbob, or the repository ZIP.
08 The agent asks to connect company systems.	Say no: the workshop uses practice data. It is in every setup prompt.
09 A run is slow or wanders.	Ask for the plan first, then run one sub-agent at a time; cut the scope to one TA, one HCP, one output.
10 The output is generic.	Teach the agent: add 3–5 house rules (what good looks like, what must never appear) and run again.



Unblock, then step back. Teams learn most when they make the call at each gate themselves.



Impact worksheet and setup prompt

Impact worksheet the team's own estimates

One cycle = one manuscript-to-abstract-to-lay-summary consistency cycle

Today baseline	h hours per cycle (all people)	×	 cycles per year	×	\$ /h loaded rate	=	\$ cost per year
vs							
AI-assisted with the swarm	h hours per cycle, incl. human review	×	 cycles per year	×	\$ /h loaded rate	=	\$ cost per year
HOURS SAVED PER YEAR (today h – AI h) × cycles =		VALUE PER YEAR (ESTIMATE) hours saved × rate =		OTHER BENEFITS			
h		\$		days speed to decision % coverage # errors caught			

ILLUSTRATIVE

Not data, only an example: 24 h → 6 h per cycle × 10 cycles/year = 180 h saved; × \$150/h = \$27,000 per year. Replace every number with the team's own estimate.

Count every person involved and include human review time in the AI-assisted row. These numbers go on the Impact slide of the final template (slide 9: formula tiles and a bar chart built from the team's own estimates).

Setup prompt Block 1 · 2:02 · the driver pastes it into Grok Bot (replace the TA)

1 · Setup prompt

copy all

Use <https://github.com/Open-Medical-Affairs/Medical-Affairs-Skills>
 Read AGENTS.md. We are the Publications Brain team at AI in Action for Medical Affairs.
 Load medical-affairs-foundations and these skills: scientific-communication-strategy, evidence-gap-analysis, scientific-manuscript, congress-abstract-and-poster, plain-language-summary, citation-integrity, deliverable-quality-review.
 Use the practice data in <https://github.com/Open-Medical-Affairs/Data-Sources>: synthetic/[oncology-mm | immunology-ad | cardiometabolic-obesity]/ and synthetic/connected/ (the practice CRM). It is fictional (Nordvant Biopharma).
 First, only: list the skills you loaded, then show the first 5 rows of content_assets.csv and confirm the file is labelled SYNTHETIC. Do not start a mission yet.
 Do not ask me to connect company systems. Do not send, post or change anything outside this conversation.

SUGGESTED ROLES (ROTATE IF YOU LIKE)

Driver

types into Grok Bot

Navigator

reads the mission and pushes back

Scribe

owns the capture log

Timekeeper + presenter

keeps the clock, tells the story



Assignment starter for the hack

Paste into Grok Bot at the start of Block 4 (3:40), then edit the [brackets]. The setup prompt for Block 1 is on page 8.
Ready prompts for every mission: aiinaction.up.railway.app/prompts · swarm mode and the Skill creator: [/optimizer](#).

2 · Assignment starter: Publications Brain swarm

Block 4 · 3:40

```
# Assignment: Publications Brain swarm
You are the Publications Strategist, a lead agent coordinating a small team of digital workers for a Medical
Affairs team. This is an assignment, not a question: plan, delegate, check and hand back finished drafts.

## Goal: what done looks like
[One sentence: the problem we picked in ideation, and the deliverable that solves it.]

## Context to use
- https://github.com/Open-Medical-Affairs/Medical-Affairs-Skills : read AGENTS.md, load medical-affairs-
foundations, your lead skill scientific-communication-strategy, and the skills named below.
- Practice data (fictional): https://github.com/Open-Medical-Affairs/Data-Sources synthetic/[TA]/ and
synthetic/connected/. Or our own non-confidential data: [describe].

## How to work
1. Show me a short plan and the org chart of workers first. Wait for my OK.
2. Scan human-sourced records for possible safety findings (adverse events, product complaints, off-label
use) before anything else. List them verbatim and stop until I acknowledge.
3. Run these sub-agents, each with only the context it needs:
  - Demand Reader (skills: field-insight-synthesis): Reads field observations and MI enquiries; lists the
questions each audience is trying to answer. Hands off: Audience-question list.
  - Evidence Mapper (skills: evidence-synthesis + evidence-gap-analysis): Maps each question to the
evidence; labels evidence gap vs communication gap. Hands off: Gap table with sources.
  - Output Drafter (skills: scientific-manuscript / congress-abstract-and-poster / plain-language-summary):
Drafts the chosen output(s) from the approved source. Hands off: Draft outputs.
  - Consistency Auditor (skills: citation-integrity + data-visualization-for-medical): Checks every number,
figure and claim against the evidence table. Hands off: Discrepancy log.
4. STOP at Gate 1: After the Evidence Mapper: the publications lead (with the steering committee) approves
the publish / stop / research list before any drafting starts. Show me the evidence pack and wait.
5. Continue, then have an independent checker (deliverable-quality-review (reports to the human, not the
lead); add mlr-review-readiness for a pre-MLR pass) review the draft. Fix what it finds.
6. STOP at Gate 2: After the Consistency Auditor: authors and the medical reviewer resolve the discrepancy
log and approve the drafts before anything goes to authors, MLR or a journal. Wait for my decision.
7. Keep a timing note: minutes per step, and minutes of human review at each gate.

## House rules
- Never invent a citation, number, quote or meeting fact. Unsupported claims are marked and left for a human.
- Keep source IDs on every claim. Keep public evidence separate from fictional practice data.
- Do not send, post, publish or change any external system. Drafts only. Mark every output DRAFT and the data
FICTIONAL.
```

All products, people and results in the practice data are fictional. Outputs are drafts that need qualified human review; this is not a clinical decision, GxP or pharmacovigilance system.

